HARD PROBLEM-SOLVING AND DATA ANALYSIS

Ratios, rates, proportional relationships, and units

02

7 questions · ~11 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

If $\frac{4a}{b} = 6.7$ and $\frac{a}{bn} = 26.8$, what is the value of n?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2 PROBLEM-SOLVING AND DATA ANALYSIS

In a scale model of an office building, a length of 1 inch is equivalent to a length of $\frac{17}{3}$ feet in the actual office building. If a length in the actual office building measures x feet, which expression represents the corresponding length, in inches, in the model in terms of x?

A. $\frac{3x}{1712}$

B. $\frac{17x}{123}$

C. $\frac{3x}{17}$

D. $\frac{17x}{3}$

Q3 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

For an electric field passing through a flat surface perpendicular to it, the electric flux of the electric field through the surface is the product of the electric field's strength and the area of the surface. A certain flat surface consists of two adjacent squares, where the side length, in meters, of the larger square is 3 times the side length, in meters, of the smaller square. An electric field with strength 29.00 volts per meter passes uniformly through this surface, which is perpendicular to the electric field. If the total electric flux of the electric field through this surface is $4{,}640$ $\text{volts} \cdot \text{meters}$, what is the electric flux, in $\text{volts} \cdot \text{meters}$, of the electric field through the larger square?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

An area of 46.00 square nautical miles is equivalent to k square kilometers. To the nearest tenth, what is the value of k? $1 \text{ nautical mile} = 1.852 \text{ kilometers}$

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5 PROBLEM-SOLVING AND DATA ANALYSIS

A certain town has an area of 4.36 square miles. What is the area, in square yards, of this town? $1 \text{ mile} = 1{,}760 \text{ yards}$

A. 404

B. 7,674

C. 710,459

D. 13,505,536

Q6 PROBLEM-SOLVING AND DATA ANALYSIS

A certain park has an area of 11,863,808 square yards. What is the area, in square miles, of this park? $1 \text{ mile} = 1{,}760 \text{ yards}$

A. 1.96

B. 3.83

C. 3,444.39

D. 6,740.8

Q7 PROBLEM-SOLVING AND DATA ANALYSIS

The speed of a vehicle is increasing at a rate of 7.3 meters per second squared. What is this rate, in **miles per minute squared**, rounded to the nearest tenth?

Use 1 mile $= 1{,}609$ meters.

A. 0.3

B. 16.3

C. 195.8

D. 220.4